RICE SOCIAL SCIENCES
Department of
Sport Management

Course: SMGT 435/535 – Baseball Analytics
Term: Fall 2026
Class: 1:00 p.m. – 2:15 p.m. Tue/Thu

Instructor Contact Information

Instructor: Scott Powers
Office: Kraft 246
Email: scott.powers@rice.edu
Office Hours: saberpowers.com/bookme

Required Texts and Materials

Class Website: canvas.rice.edu
Required Text: All readings will be made available on Canvas.
Required Material: Bring your laptop to class every day, but please leave your laptop packed away when we are not working on coding exercises.

Topics

The general flow is that we will alternate between (a) discussing math/models using a whiteboard and (b) implementing these concepts in code. The course is divided into 4 units, each approximately 3 weeks:

- Unit 1: Measuring Batting Performance
  Pythagorean Formula, Base-Out Run Expectancy and Linear Weights, Batted Ball Outcome Model
- Unit 2: Measuring Pitching Performance
  BABIP/FIP/DIPS, Pitch-Level Analysis, Pitch Outcome Modeling
- Unit 3: Advanced Topics
  "Stuff", Fielding and Baserunning, Projections
- Unit 4: Practicum
  Guest Speakers, Student Projects

Course Objectives and Learning Outcomes

After successfully completing this course, you will be able to:

- Separate signal from noise in baseball performance data.
- Design and calculate your own WAR metric using R and publicly available MLB data.
- Describe applications of analytics for supporting in-game strategy and player development.
- Identify an interesting research problem in baseball, solve it with data, and present your solution.

How Your Grade Is Determined

Sport analytics is a demanding and highly competitive field. This class aims to provide the tough but honest feedback that you will need to succeed in it. Note that the points below sum approximately (but not exactly) to 100. Your grade is based on your percentage earned of available points.

			$[93\%, \infty)$	$\rightarrow$	A	$[90\%, 93\%)$	$\rightarrow$	A–
$[87\%, 90\%)$	$\rightarrow$	B+	$[83\%, 87\%)$	$\rightarrow$	B	$[80\%, 83\%)$	$\rightarrow$	B–
$[77\%, 80\%)$	$\rightarrow$	C+	$[73\%, 77\%)$	$\rightarrow$	C	$[70\%, 73\%)$	$\rightarrow$	C–
$[67\%, 70\%)$	$\rightarrow$	D+	$[63\%, 67\%)$	$\rightarrow$	D	$[60\%, 63\%)$	$\rightarrow$	D–
			$[0\%, 60\%)$	$\rightarrow$	F			

Assignments. (45pts) There are 3 individual assignments, each worth 15pts. You are encouraged to help each other, but your submission must reflect work you have done yourself. Assignments will be graded anonymously, so please remove any personally identifying information from your submission. The expected completion time for each assignment is 15 hours.

Project. (45pts) This is a project-based course, and the project you build in stages over the course of the semester will determine a large chunk of your grade. You are encouraged to work with a partner on the project (no more than 2 students per team), but you may work individually if you wish.

- Project #0 (0pts): Registration — expected hours: 0
- Project #1 (5pts): Proposal — expected hours: 5
- Project #2 (10pts): Abstract — expected hours: 10
- Project #3 (10pts): Presentation — expected hours: 10
- Project #4 (20pts): Paper — expected hours: 20

Attendance. (1pt per class) Showing up is a requirement for most jobs in sport analytics. In this class, you are expected to show up and engage in discussions, which are part of the learning experience. We will keep attendance and collect feedback using exit tickets at the end of almost every class although there may be a few days we skip this (these days will not be counted toward your attendance grade).

Policies

Attendance Policy. If you notify me via email at least one hour before class that you will be absent, then I will excuse you from completing the exit ticket. Please do not expect a response to your email. It is your responsibility to check the schedule and educate yourself on the material you missed.

Generative AI Policy. Generative AI may hinder your learning when used irresponsibly. To that end:

1. You may use generative AI to help you solve coding problems, but it is a violation of the honor code to submit AI-generated code that you do not understand or cannot explain.
2. It is a violation of the honor code to submit AI-generated writing as your own work.

Late Policy. Late submissions are subject to 10% deduction per day.

Tentative Schedule

Assignment Due	Date	Lesson Plan
	Aug 25	Pythagorean Formula Discussion
	Aug 27	Pythagorean Formula R Tutorial
	Sep 1	Base-Out Run Expectancy and Linear Weights Discussion
	Sep 3	Base-Out Run Expectancy and Linear Weights R Tutorial
	Sep 8	Batted Ball Outcome Model Discussion
	Sep 10	Batted Ball Outcome Model R Tutorial
	Sep 15	BABIP, FIP and DIPS Discussion
Assignment #1: Signal v. Noise in Batter Outcomes	Sep 17	BABIP, FIP and DIPS R Tutorial
	Sep 22	
	Sep 24	**NO CLASS (POWERS TRAVEL)**
	Sep 29	Pitch-Level Analysis Discussion
	Oct 1	Pitch-Level Analysis R Tutorial
	Oct 6	Pitch Outcome Model Discussion
Project #0: Registration	Oct 8	Pitch Outcome Model R Tutorial
	Oct 13	**NO CLASS (MIDTERM RECESS)**
Assignment #2: Pitch Outcome Model	Oct 15	**NO CLASS (POWERS TRAVEL)**
	Oct 20	"Stuff" Discussion
	Oct 22	"Stuff" R Tutorial
	Oct 27	Fielding and Baserunning Discussion
	Oct 29	Fielding and Baserunning R Tutorial
	Nov 3	Projections Discussion
Project #1: Proposal	Nov 5	Projections R Tutorial
	Nov 10	Guest Speaker (Diego Martinez)
Assignment #3: Projections	Nov 12	Guest Speaker (Ryan Ferguson)
	Nov 17	
Project #2: Abstract	Nov 19	
	Nov 24	Guest Speaker (Perla Villarreal)
	Nov 25	**NO CLASS (THANKSGIVING RECESS)**
Project #3: Presentation	Dec 1	Student Presentations
	Dec 3	Student Presentations
Project #4: Paper	Dec 14	

Experiential Learning

MLB Club Interview Assessment. If you are asked to complete a modeling assessment during an interview process with an MLB club (see saberpowers.com/jobs), you are invited to substitute your work on that assessment in place of any individual homework assignment if: (1) the assessment involves coding, not just a questionnaire; (2) you notify me of this substitution before the deadline of the substituted assignment; and (3) you submit the prompt and your work on Canvas so that I may assess your work.

MLB Club Project Partnership. We will partner with one more more MLB clubs to provide mentorship on your projects. You are expected to coordinate with your mentor(s) to schedule a 30-minute meeting after each stage of your project (five total meetings) to solicit their feedback on your work.

Rice Honor Code

In this course, all students will be held to the standards of the Rice Honor Code, a code that you pledged to honor when you matriculated at this institution. If you are unfamiliar with the details of this code and how it is administered, you should consult the Honor System Handbook at `http://honor.rice.edu/honor-system-handbook/`. This handbook outlines the University's expectations for the integrity of your academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process.

Disability Resource Center

The Americans with Disabilities Act ensures that all qualified individuals have equal access to educational opportunities, regardless of disability. Students with a documented disability who require academic accommodations should contact the Student Disability Resource Center (sdrc.rice.edu) to submit appropriate documentation or confirm that their records are current. The SDRC is located in Allen Center 111 and can be reached at sdrc@rice.edu or x5841.

For additional information regarding access and accommodations for individuals with disabilities, please review Rice University Policy 402: policy.rice.edu/402.

Mental Health Policy

The wellbeing and mental health of students is important; if you are having trouble completing your coursework, please reach out to the Wellbeing and Counseling Center. Rice University provides cost-free mental health services through the Wellbeing and Counseling Center to help you manage personal challenges that threaten your personal or academic well-being. If you believe you are experiencing unusual amounts of stress, sadness, or anxiety, the Student Wellbeing Office or the Rice Counseling Center may be able to assist you. The Wellbeing and Counseling Center is located in the Gibbs Wellness Center and can be reached at 713-348-3311 (available 24/7).

Title IX Responsible Employee Notification

At Rice University, unlawful discrimination in any form, including sexual misconduct, is prohibited under Rice Policy on Harassment and Sexual Harassment (Policy 830) and the Student Code of Conduct.

Please be aware that all employees of Rice University are "mandatory reporters", which means that if you tell me about a situation involving sexual harassment, sexual assault, dating violence, domestic violence, or stalking, I must share that information with the Title IX Coordinator.

Although I have to make that notification, you will control how your case will be handled, including whether or not you wish to pursue a formal complaint. Our goal is to make sure you are aware of the range of options available to you and have access to the resources you need.

To report sexual harassment, please contact the Title IX Coordinator at titleix@rice.edu. To explore supportive measures and other resources that are available to you, please visit the Office if Interpersonal Misconduct Prevention and Support at safe.rice.edu.

This syllabus is only a guide for the course and is subject to change with advance notice.